



RECALL INDEX PROCESSES



A perspective of PT where we swap parameters instead of states.

► $Y_t^n \in X$ Is the state stored on the machine n at time t

► $\beta_t^n \in \{\beta_0, \dots, \beta_N\}$ the annealing parameter

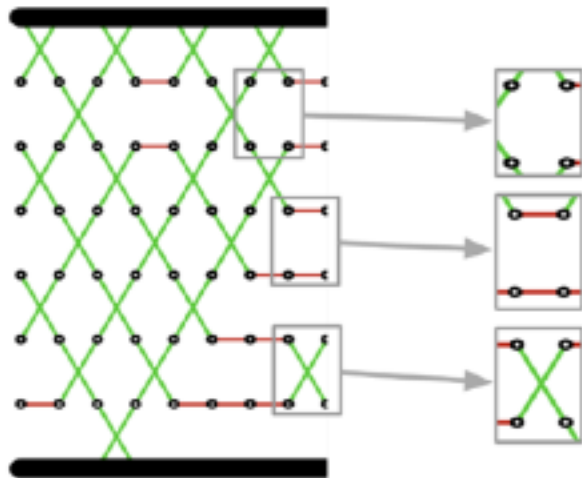
► $e_t^n \in \{-1, 1\}$ encodes the proposed direction

► We define the index process (I_t^n, e_t^n) for machine n to encode the trajectory of annealing



I_t^n denotes the index of $\beta_t^n := \beta_{I_t^n}$

► We will drop the m when it is not relevant



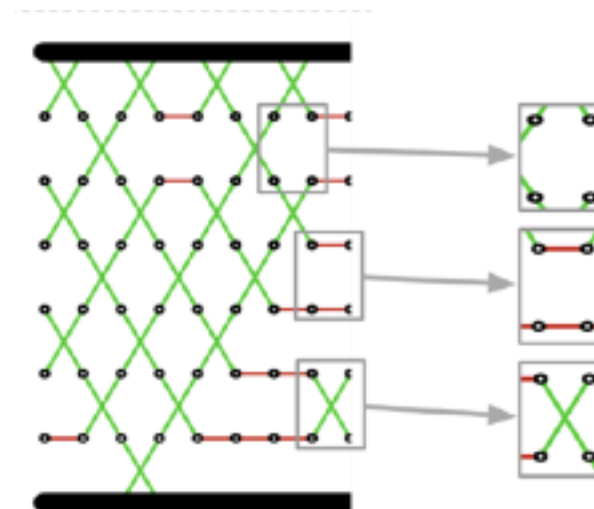
► The n -th **replica** refers to the information stored on machine n :

► The n -th **chain** refers to X_t^n , the n -th component of the PT chain $\mathbf{X}_t$ targeting π_{β_n}

► Classically the communication in PT swaps states

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- ▶ Classically the communication in PT swaps states
 - ▶ The n -th **chain** refers to X_t^n , the n -th component of the PT chain $\mathbf{X}_t$ targeting π_{β_n}
- ▶ A perspective of PT where we swap parameters across machines instead of states.
 - ▶ The m -th **replica** refers to the information stored on machine m :
 - ▶ $Y_t^m \in \mathbb{X}$ is the state stored on the machine m at time t
 - ▶ $\beta_t^m \in \{\beta_0, \dots, \beta_N\}$ the annealing parameter
 - ▶ $\epsilon_t^m \in \{-1, 1\}$ encodes the proposed direction
- ▶ We define the index process (I_t^m, ϵ_t^m) for machine m to encode the trajectory of annealing
 - ▶ I_t^m denotes the index of $\beta_t^m := \beta_{I_t^m}$
 - ▶ We will drop the m when it is not relevant



ROUND TRIPS AND RESTARTS